

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

If $|4x - 4| = 112$, what is the positive value of $x - 1$?

Correct Answer: 28

Rationale

The correct answer is **28**. The given absolute value equation can be rewritten as two linear equations: $4x - 4 = 112$ and $-(4x - 4) = 112$, or $4x - 4 = -112$. Adding **4** to both sides of the equation $4x - 4 = 112$ results in $4x = 116$. Dividing both sides of this equation by **4** results in $x = 29$. Adding **4** to both sides of the equation $4x - 4 = -112$ results in $4x = -108$. Dividing both sides of this equation by **4** results in $x = -27$. Therefore, the two values of $x - 1$ are $29 - 1$, or **28**, and $-27 - 1$, or **-28**. Thus, the positive value of $x - 1$ is **28**.

Alternate approach: The given equation can be rewritten as $|4(x - 1)| = 112$, which is equivalent to $4|x - 1| = 112$. Dividing both sides of this equation by **4** yields $|x - 1| = 28$. This equation can be rewritten as two linear equations: $x - 1 = 28$ and $-(x - 1) = 28$, or $x - 1 = -28$. Therefore, the positive value of $x - 1$ is **28**.